

## Format of Question paper for Midsemester Examination

**Birla Institute of Technology & Science, Pilani**  
**Work Integrated Learning Programmes Division**  
**Second Semester 2023-2024**

**Mid Semester Examination**  
**(EC-2 Makeup)**

Course No. : AIMLCL ZG517  
Course Title : FAccT Machine Learning  
Nature of Exam : Closed Book  
Weightage : 30%  
Duration :  
Date of Exam :

|                  |     |
|------------------|-----|
| No. of Pages     | = 2 |
| No. of Questions | = 3 |

**Note to Students:**

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1.

[1+3+2+3=9 marks]

- A. What benefit does optimized preprocessing bring in over different types of sampling based preprocessing techniques?

To turn the manual resampling process into an optimization based approach, adding more criteria beyond demographic parity and addressing the requirement of individual fairness.

- B. What are the criteria that are jointly optimized in optimized preprocessing technique? Maximize a utility function while minimizing the distortion and discrimination control.

$$\min_{p_{\hat{X}, \hat{Y}|X, Y, D}} \Delta(p_{\hat{X}, \hat{Y}}, p_{X, Y})$$

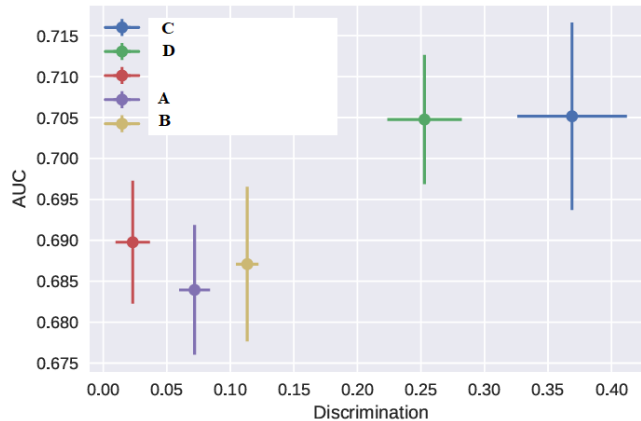
$$\text{s.t. } J(p_{\hat{Y}|D}(y|d), p_{Y_T}(y)) \leq \epsilon_{y,d} \text{ and}$$

$$\mathbb{E}[\delta((x, y), (\hat{X}, \hat{Y})) | D = d, X = x, Y = y] \leq c_{d,x,y}$$

- C. What is the nature of the resulting optimization problem?

Quasi convex optimization with inequality constraints.

- D. Match A, B, C, D in the following plot with logistic regression with all parameters, logistic regression without sensitive attribute, optimized preprocessing with distortion control parameter  $\epsilon = 0.1$  and  $\epsilon = 0.25$ . Explain your answer clearly.



C = Logistic regression with all parameters D = LR w/o sensitive attribute, A= optimized with  $\epsilon = 0.25$  and B= optimized with  $\epsilon = 0.1$ . Since sensitive attribute is used in D, discrimination is higher than C which does not use sensitive attribute. A & B both have lower discrimination as expected than A & B but their accuracy (AUC) is also lower. Among A & B, A uses a higher value of  $\epsilon$  thus resulting in lower discrimination at the expense of accuracy w.r.t. B.

Q.2.

[4+1+2+4+2+1=14 Marks]

Consider the training dataset D given in the Table 1. Sex is the only protected attribute.

A. Calculate the prejudice index and normalized prejudice index of D.

$$\begin{aligned} P(y=+) &= 0.6 & P(-) &= 0.4 & P(M) &= P(F) = 0.5 \\ P(+, M) &= 0.4 & P(-, M) &= 0.1 & P(+, F) &= 2/10 & P(-, F) &= 3/10 \end{aligned}$$

$$\begin{aligned} PI &= P(+, M) \ln(\Pr(+, M) / [\Pr(+)\Pr(M)]) + P(-, M) \ln(\Pr(-, M) / [\Pr(-)\Pr(M)]) + \\ &\quad P(-, F) \ln(\Pr(-, F) / [\Pr(-)\Pr(F)]) + P(+, F) \ln(\Pr(+, F) / [\Pr(+)\Pr(F)]) \\ &= 0.4 \ln(0.4/0.3) + 0.1 \ln(0.1/0.2) + 0.3 \ln(0.3/0.2) + 0.2 \ln(0.2/0.3) \\ &= 0.0863 \end{aligned}$$

$$NPI = 0.0863 / \sqrt{1 * (0.6 \ln(0.6) + 0.4 \ln(0.4))} = 0.105$$

B. Use one hot encoding to represent the categorical variables, and design and draw a logistic regression classifier. Clearly show how many inputs and outputs will you require?

If 0/1 encoding is used, 4 inputs are used and 1 output. If 01 / 10 encoding is used, 8 inputs will be used.

| Sex | Ethnicity | Highest degree | Job type   | Class |
|-----|-----------|----------------|------------|-------|
| M   | Native    | H. school      | Healthcare | +     |
| M   | Native    | Univ.          | Healthcare | +     |
| M   | Native    | H. school      | Healthcare | +     |
| M   | Non-nat.  | H. school      | Healthcare | +     |
| M   | Non-nat.  | Univ.          | Healthcare | -     |
| F   | Non-nat.  | Univ.          | Education  | -     |
| F   | Native    | H. school      | Education  | -     |
| F   | Native    | H. school      | Healthcare | +     |
| F   | Non-nat.  | Univ.          | Education  | -     |
| F   | Native    | H. school      | Healthcare | +     |

**Table 1. Sample relation for the job-application example**

C. The logistic regression based classifier designed in B with prejudice removal regularizer is trained to obtain fair decisions. All classifier parameters except the one associated with Gender are initialized to 0. Parameter  $w_{\text{gender}}$  associated with Gender is initialized to 1. Assuming

- Without prejudice removal regularizer,
- with the prejudice removal regularizer. Show all calculation steps clearly. Assume prediction=1 if actual output  $\geq 0.5$ , else 0.

Let us represent M/F by 0/1, Native/Non- learning rate = 0.1, regularization constant = 0.5, and full batch training, calculate the value of classifier parameters after 1<sup>st</sup> iteration without the prejudice removal native by 0/1, H. School/Univ by 0/1 and Education/Healthcare is 0/1. Thus,  $w_0=w_{\text{ethnic}}=w_{\text{deg}}=w_{\text{job}}=0$ .

- i) So, first five outputs will be 0.5 and last 5 outputs will be  $1/(1+\exp(-1))$
- $$w_0(1)=w_0(0)-0.1/10*(4*(0.5-1)+0.5+3*0.73+2*(0.73-1))=-0.0015$$
- $$w_{\text{gender}}(1)=1.0-0.1/10*(4*(0.5-1)*0+0.5*0+3*0.73+2*(0.73-1)*1)=0.9835$$
- $$w_{\text{ethnic}}(1)=-0.1/10*(3*(0.5-1)*0+0.5*1+3*0.73*1)=-0.0269$$
- $$w_{\text{degree}}(1)=-0.1/10*(1*(0.5-1)+3*0.73*1)=-0.0169$$
- $$w_{\text{job}}(1)=-0.1/10*(3*0.73*1)=-0.0219$$

- ii) If PI removal regularization is used, regularization loss R to be optimized is given by

$$\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

$$\hat{\text{Pr}}[y] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}]}{|\mathcal{D}|}$$

$$\hat{\text{Pr}}[y|s] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i=s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i=s\}|}$$

| Sex | Ethnicity | Degree | Job Type | Class | Actual Output $\mathcal{M}[y \mathbf{x},s,0]$ |
|-----|-----------|--------|----------|-------|-----------------------------------------------|
| 0   | 0         | 0      | 0        | 1     | 0.5                                           |
| 0   | 0         | 1      | 0        | 1     | 0.5                                           |
| 0   | 0         | 0      | 0        | 1     | 0.5                                           |
| 0   | 1         | 0      | 0        | 1     | 0.5                                           |
| 0   | 1         | 1      | 0        | 0     | 0.5                                           |
| 1   | 1         | 1      | 1        | 0     | 0.731059                                      |
| 1   | 0         | 0      | 1        | 0     | 0.731059                                      |
| 1   | 0         | 0      | 0        | 1     | 0.731059                                      |
| 1   | 1         | 1      | 1        | 0     | 0.731059                                      |
| 1   | 0         | 0      | 0        | 1     | 0.731059                                      |

$$\text{Pr}[y]=0.6155$$

$$\text{Pr}[y|s=0]=0.5$$

$$\text{Pr}[y|s=1]=0.731$$

$$\ln(\text{Pr}[y|s=0]/\text{Pr}[y])=-0.208$$

$$\ln(\text{Pr}[y|s=1]/\text{Pr}[y])=0.172$$

$$w_0(1)=-0.0015$$

$$w_{\text{gender}}(1)=0.9835-0.05/10*(4*(0.5-1)*0+0.5*0+3*0.73*0.172+2*(0.73-1)*1*0.172)=0.9821$$

$$w_{\text{ethnic}}(1) = -0.0269 - 0.05/10 * (3 * (0.5 - 1) * 0 + 0.5 * 1 * -0.208 + 3 * 0.73 * 1 * 0.172) \\ = -0.02826$$

$$w_{\text{degree}}(1) = -0.0169 - 0.05/10 * (1 * (0.5 - 1) * -0.208 + 3 * 0.73 * 1 * 0.172) = -0.0193$$

$$w_{\text{job}}(1) = -0.0219 - 0.05/10 * (3 * 0.73 * 1 * 0.172) = -0.02378$$

D. Calculate the demographic parity difference before and after regularization.

Demographic parity difference before and after PR regularization after 1 iteration=0 based on actual outputs calculated with new weights.

E. What type of prejudice removal is addressed in designing prejudice removal regularization?

Indirect prejudice

Q.3.

**[3+2+2=7 Marks]**

A. Draw the L1, L2 and L3 causal models for counterfactual fairness.

Refer to slides.

B. What is multi-world counterfactual fairness? What is the rationale behind this paradigm?

A counterfactual model developed with assumptions of multiple causal models is known as multi-world counterfactual fairness. Refer to slides for the associated loss function.

There could be multiple causal graphs with latent variables associated with an observed dataset. So, multi-world counterfactual fairness has been introduced.

C. Illustrate one example of fair computer vision where counterfactual fairness is applied.

An image of a smile face is classified as smiling by a smile detector even when the facial hair is altered keeping rest of the face the exactly same (so that a male smiling face resembles a female smiling face!)